Order Statistics and their Distributions

Order Statistics

- Let $X_1, \ldots X_n$ denote a random sample from a distribution of the *continuous type* having pdf $f(x)$ on support $\mathcal{S} = (a, b)$, where $-\infty \leq a < b \leq \infty$.
- The joint pdf of $X_1, \ldots X_n$ is

$$f(x_1, \ldots, x_n) = f(x_1)f(x_2)\cdots f(x_n).$$

- **Order Statistics**: Let Y_1 be the smallest of these X_i, Y_2 be the next X_i in order of magnitude, ..., and Y_n be the largest of X_i. That is $Y_1 < Y_2 < \cdots < Y_n$ represent $X_1, \ldots, X_n$ when the latter are arranged in ascending order. We call Y_i the ith order statistics of the random sample of $X_1, \ldots, X_n$.

Marginal Distributions of Y_1 and Y_n

- Question: How to obtain the pdf of Y_1? How to obtain the pdf of Y_n?

Marginal Distributions of Y_1 and Y_n

Joint pdf of Order Statistics

- **Theorem**: The joint pdf of $Y_1, Y_2, \ldots, Y_n$ is given by

$$g(y_1, \ldots, y_n) = \begin{cases} n! f(y_1) f(y_2) \cdots f(y_n) & a < y_1 < y_2 < \cdots < y_n < b \\ 0 & \text{elsewhere} \end{cases}$$

An Example

- Continuous random variable X have pdf $f(x)$ on (a, b). Let X_1, X_2, X_3 denote a random sample from this distribution and let $Y_1 < Y_2 < Y_3$ denote the order statistics of the sample. We shall compute the probability $Y_2 < m$, where m is the median of X. The joint pdf of three is

$$g(y_1, y_2, y_3) = \begin{cases} 6f(y_1)f(y_2)f(y_3) & a < y_1 < y_2 < y_3 < b \\ 0 & \text{elsewhere} \end{cases}$$

An Example

The pdf of Y_2 is

$$
\begin{aligned}
h(y_2) &= 6f(y_2)\int_{y_2}^{b}\int_{a}^{y_2} f(y_1)f(y_3)dy_1dy_3 \\
&= 6f(y_2)F(y_2)(1-F(y_2)) \quad a < y_2 < b
\end{aligned}
$$

So,

$$
P(Y_2 \le m) = 6\int_{a}^{m}\{F(y_2)f(y_2) - [F(y_2)]^2 f(y_2)\}dy_2 = \frac{1}{2}.
$$

Marginal pdf of an Order Statistic

- It is easy to express the marginal pdf of any order statistics, say Y_k, in terms of $F(x)$ and $f(x)$.

$$g_k(y_k) = \frac{n!}{(k-1)!(n-k)!}[F(y_k)]^{k-1}(1-F(y_k))^{n-k}f(y_k)$$

for $a < y_k < b$.

- *Easy method to remember the pdf of order statistics*: The probability $P(y_i < Y_i < y_i + \Delta)$, where Δ is small, can be approximate by the following multinomial probability. In n independent trials, $i-1$ outcomes must be less than y_i[an event with probability $p_1 = F(y_i)$]; $n-i-1$ outcomes much be greater than $y_i + \Delta$ [an event with probability $p_2 = 1 - F(y_i)$]; Final one outcome must be between y_i and $y_i + \Delta$ with probability $p_3 = f(y_i)\Delta$. This multinomial probability is

$$\frac{n!}{(i-1)!(n-i)!} p_1^{i-1} p_2^{n-i} p_3$$

which is $g(y_i)\Delta$.

An Example

- Let $Y_1 < Y_2 < Y_3 < Y_4$ denote the order statistics of random sample of size 4 from pdf

$$f(x) = 2x, \quad 0 < x < 1.$$

Here $F(x) = x^2, 0 < x < 1$. So that

$$g_3(y_3) = \frac{4!}{2!1!}(y_3^2)^2(1 - y_3^2)(2y_3), \ \ 0 < y_3 < 1.$$

Thus,

$$P(Y_3 > \frac{1}{2}) = \int_{1/2}^{\infty} g_3(y_3)dy_3 = \frac{243}{256}.$$

Joint pdf of Two Order Statistics

- It is also easy to express the joint pdf of any two order statistics,say $Y_i < Y_j$ in terms of $F(x)$ and $f(x)$.

$$g_{i,j}(y_i, y_j) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!}[F(y_i)]^{i-1}[F(y_j) - F(y_i)]^{j-i-1}$$

$$\times[1 - F(y_j)]^{n-j} f(y_i) f(y_j), \quad a < y_i < y_j < b.$$